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Hara et al.

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(54) **SUCTION CUP DEVICE**

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(73) Assignee: **Sony Corporation**, Tokyo (JP)

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A47G 29/00 (2006.01)
F16B 47/00 (2006.01)
A45D 42/14 (2006.01)

(52) **U.S. Cl.** **248/205.8**; 248/205.5; 248/205.6; 248/205.7; 248/206.1; 248/206.2; 248/363

(58) **Field of Classification Search** 248/205.5, 248/205.6, 205.7, 205.8, 206.1, 206.2, 363
See application file for complete search history.

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(57) **ABSTRACT**

A suction cup device, includes: a suction cup; a center shaft; an attaching base; a rock lever; a coil spring; and a cam mechanism.

11 Claims, 9 Drawing Sheets

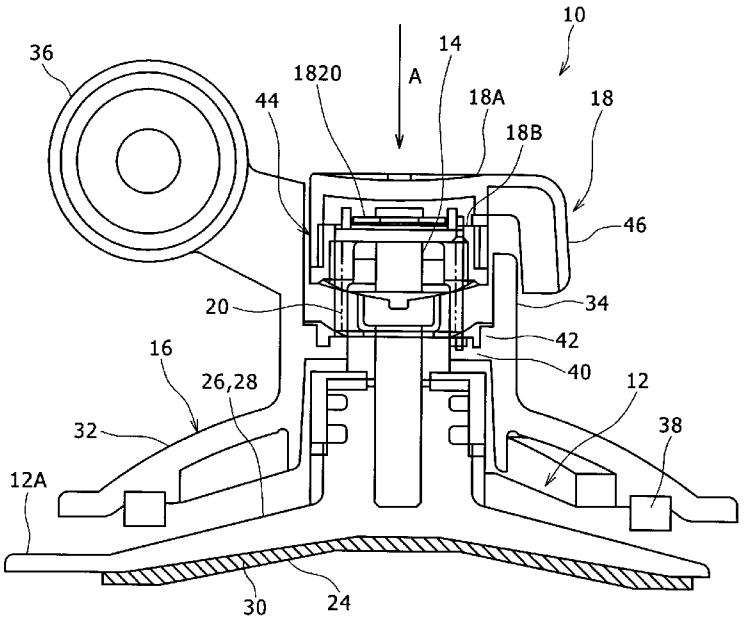


FIG. 1

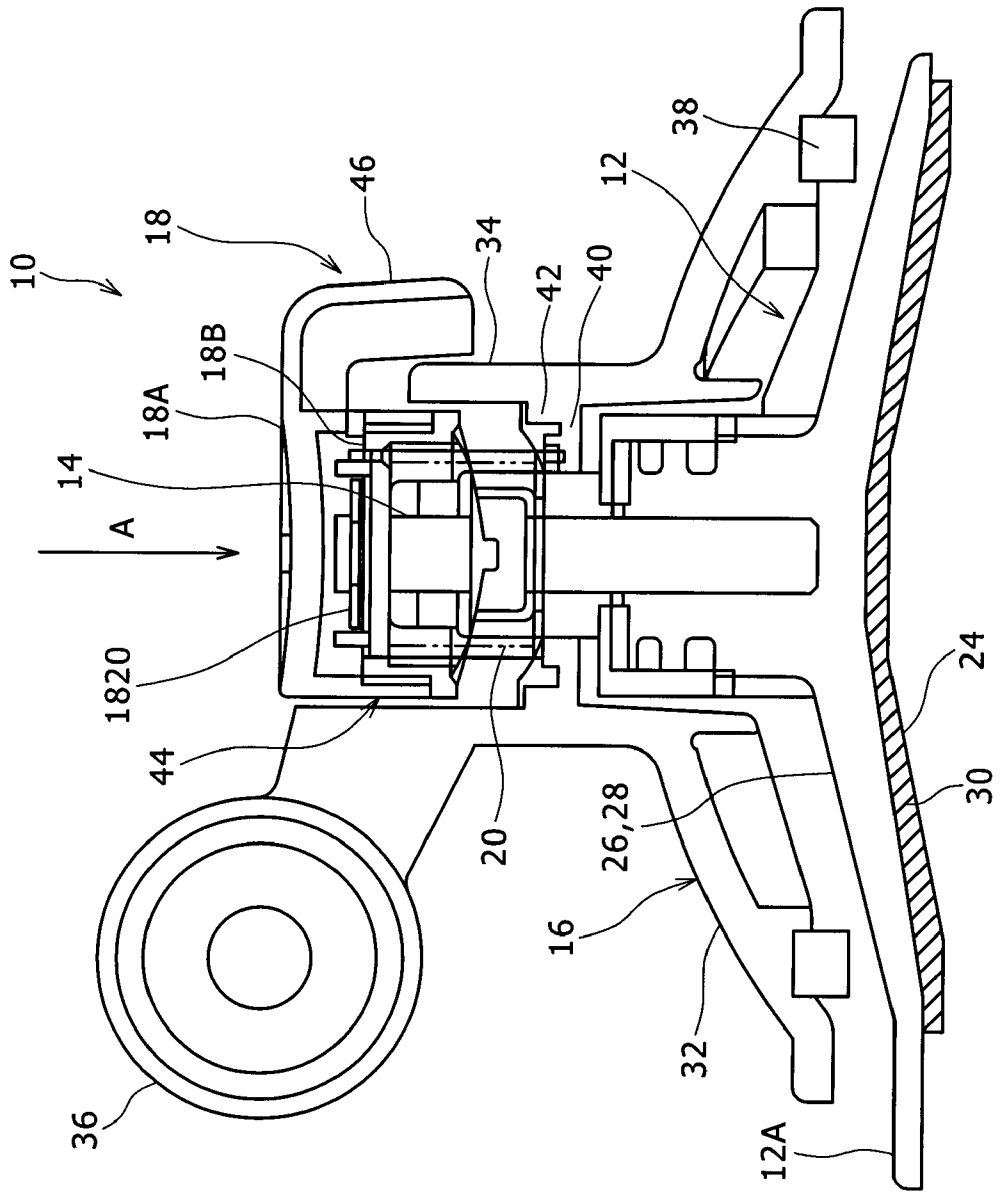

10
18
46
34
42
40
12
38
18B
18A
14
A
1820
44
20
26,28
16
32
36
12A
24
30

FIG. 2

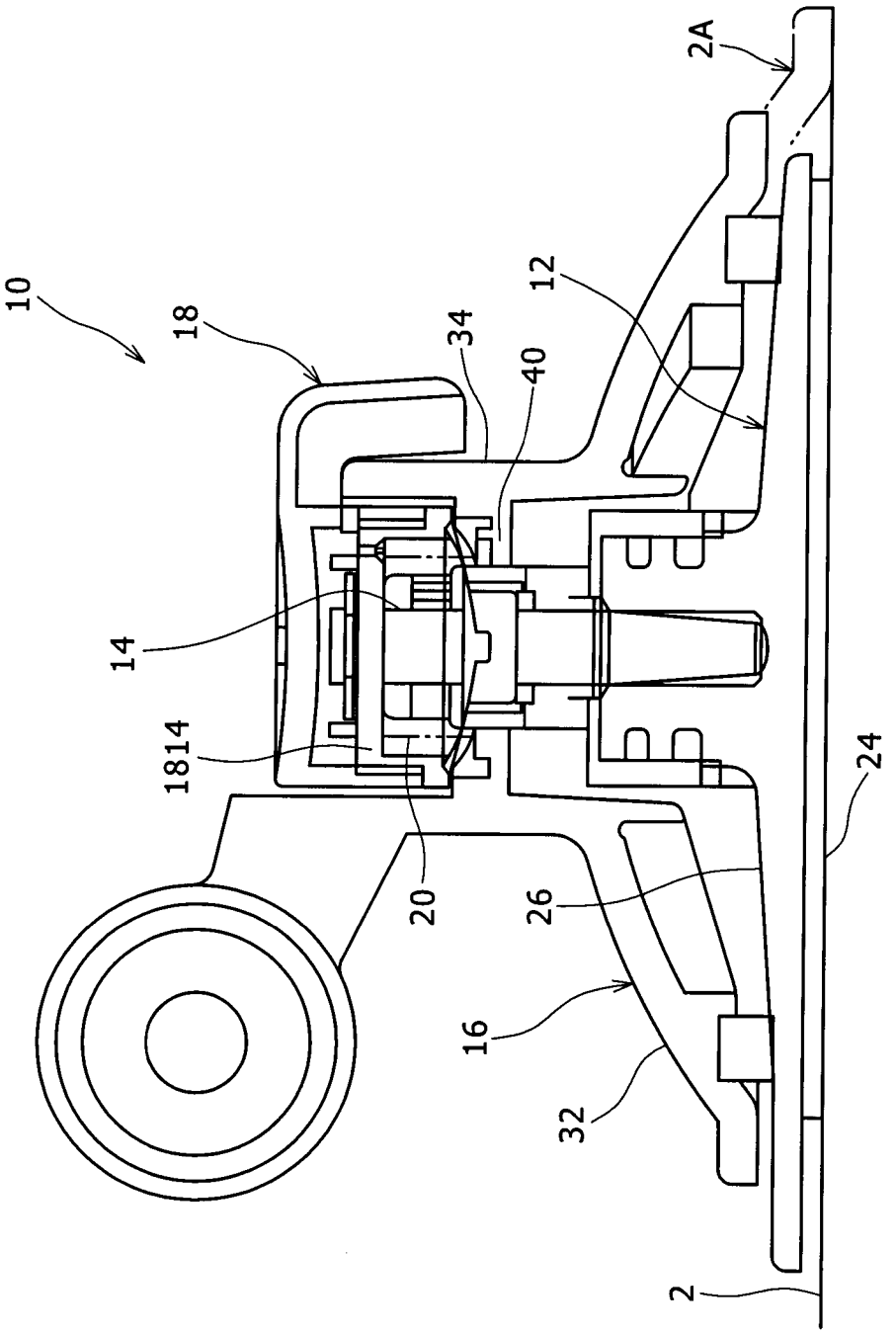


FIG. 3

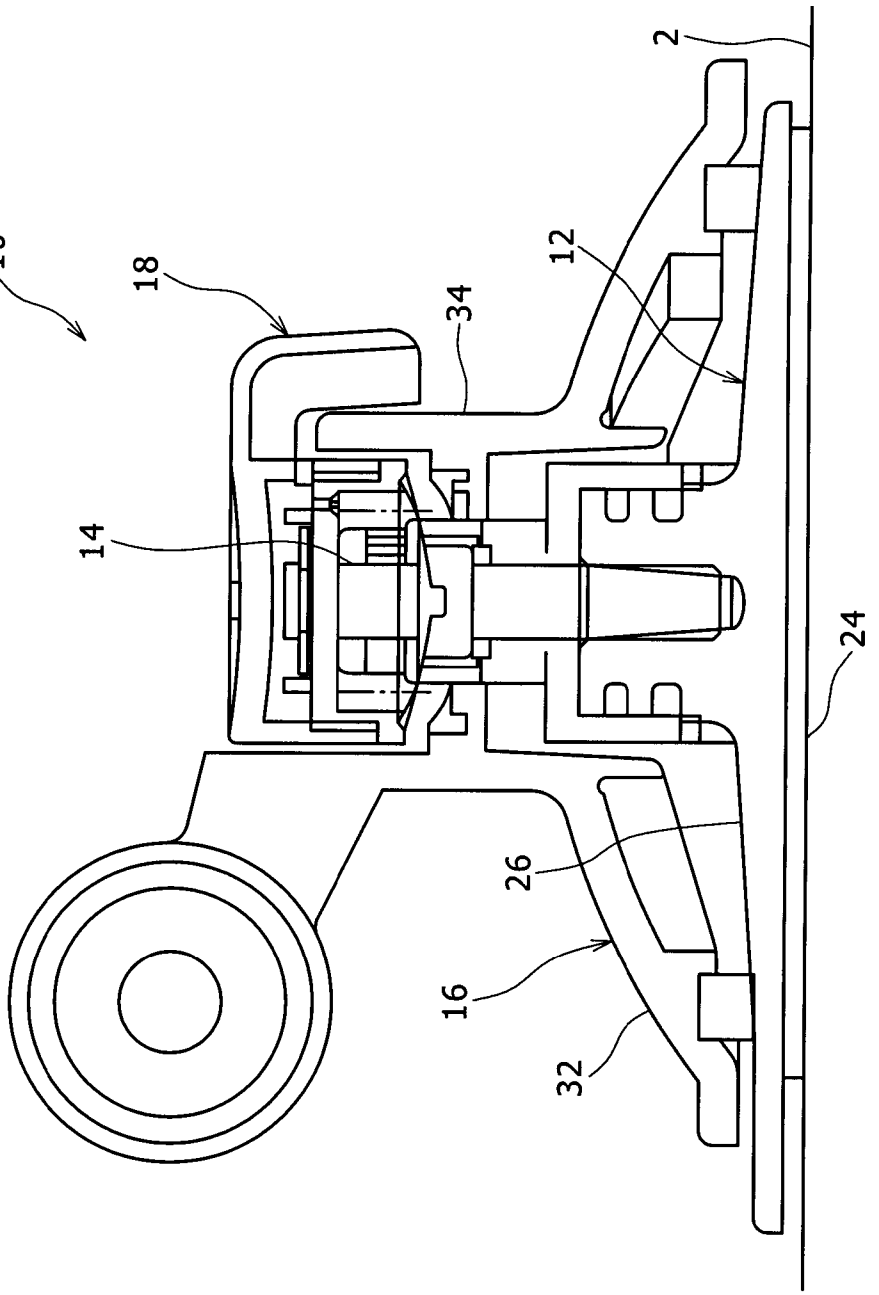


FIG. 4

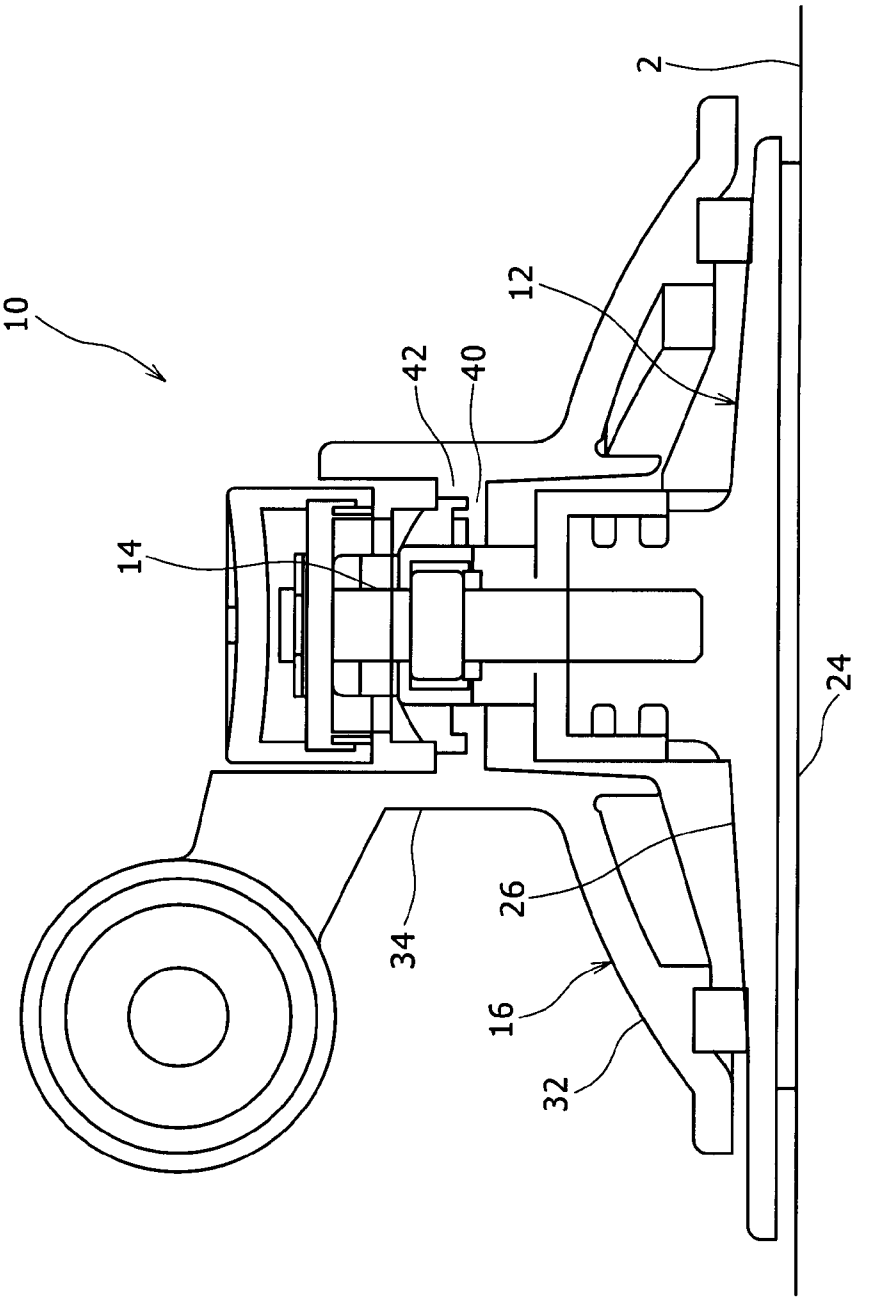


FIG. 5A

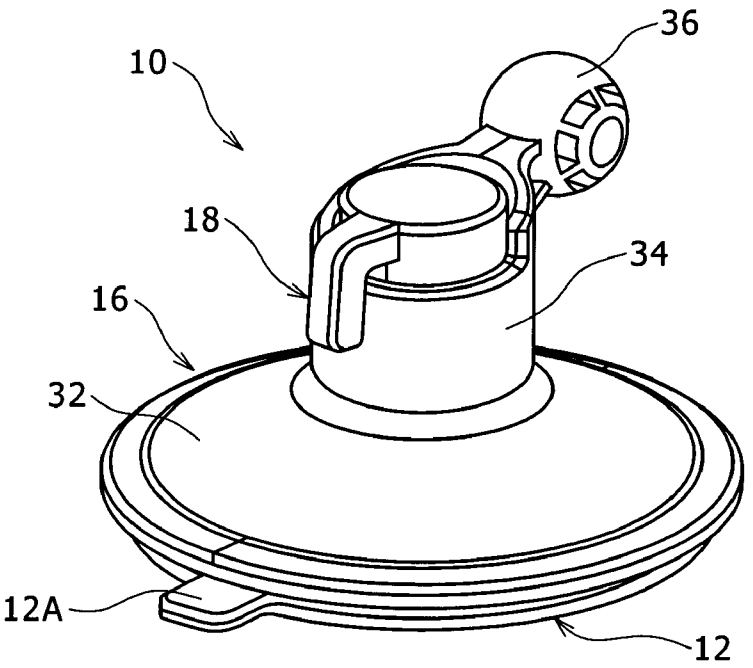


FIG. 5B

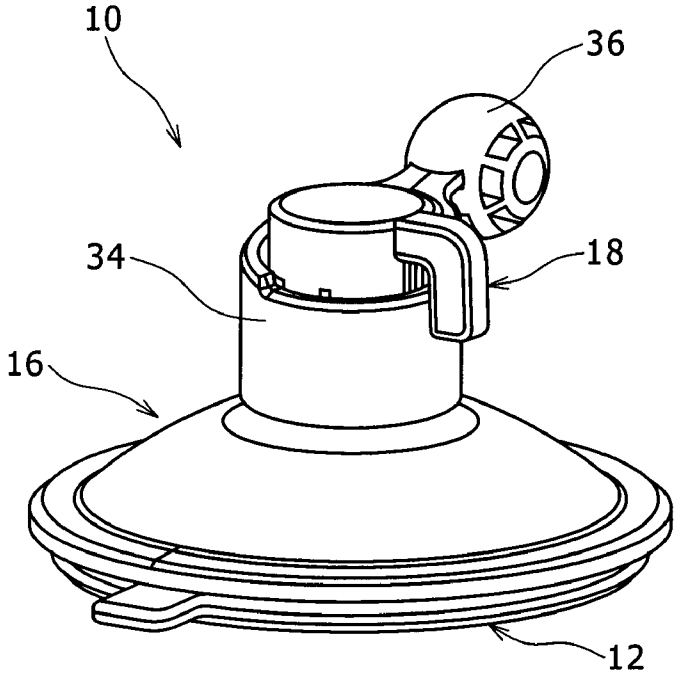


FIG. 6

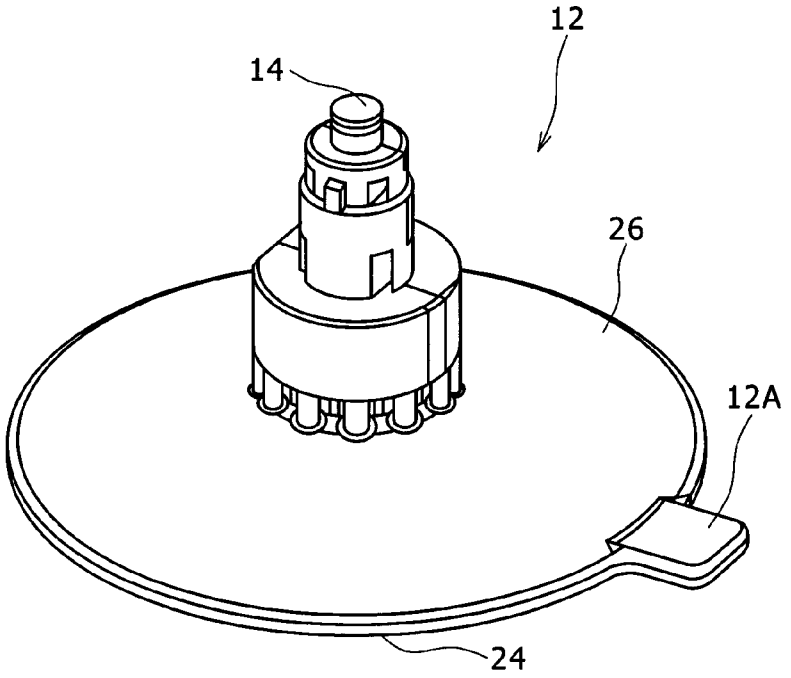


FIG. 7

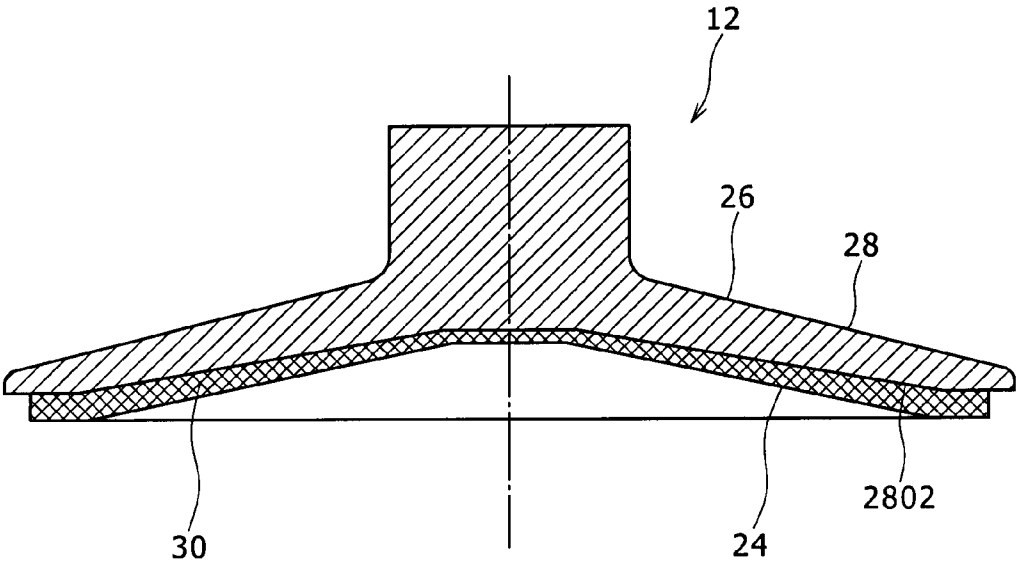


FIG. 8

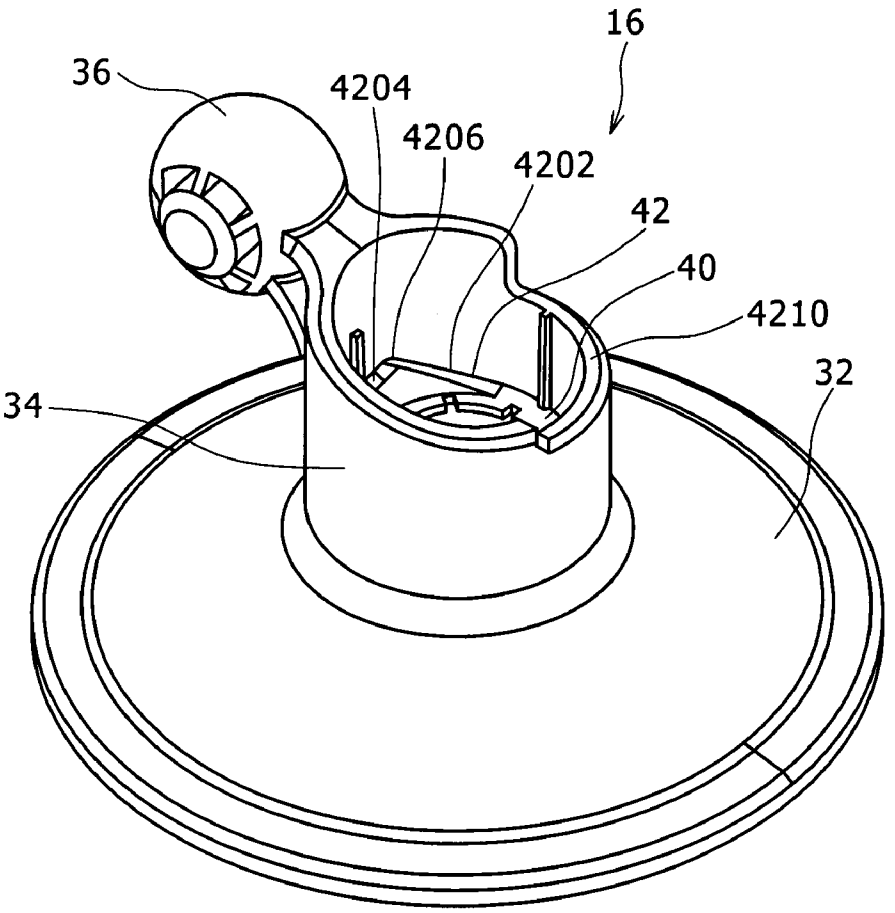


FIG. 9A

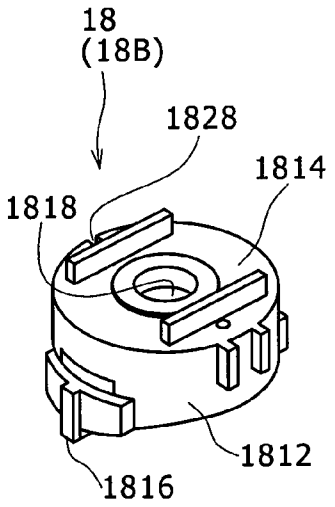


FIG. 9C

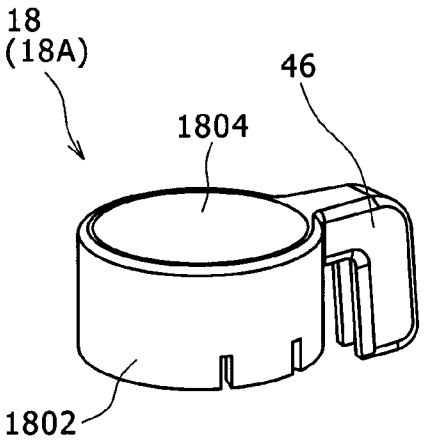


FIG. 9B

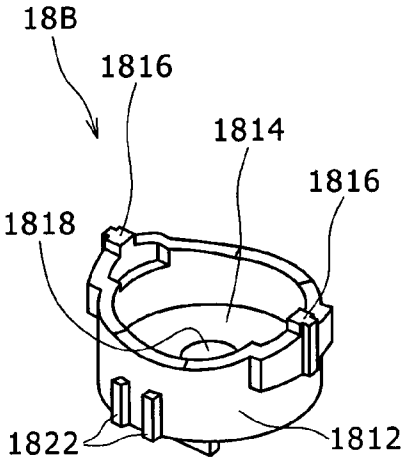


FIG. 9D

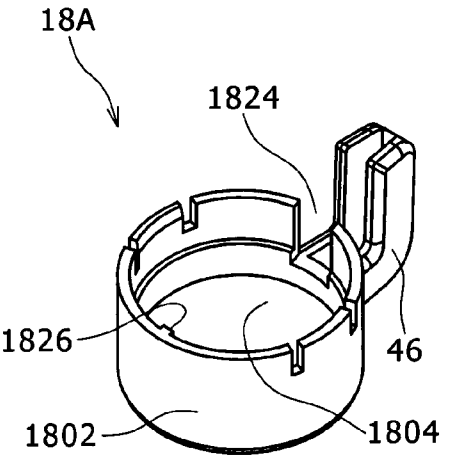


FIG. 10

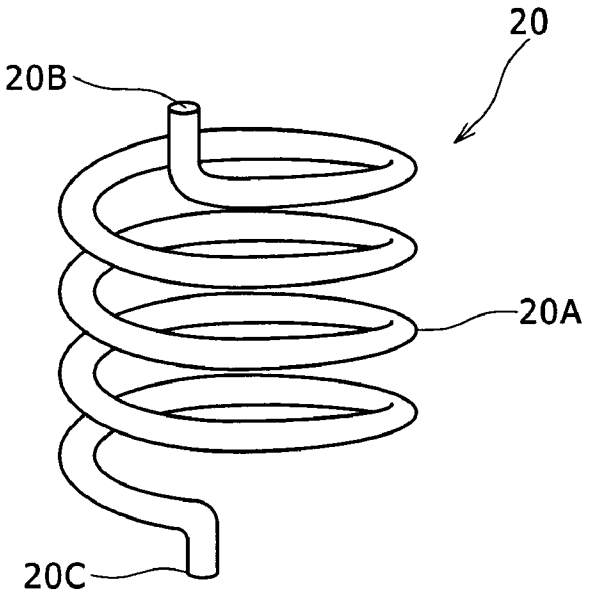
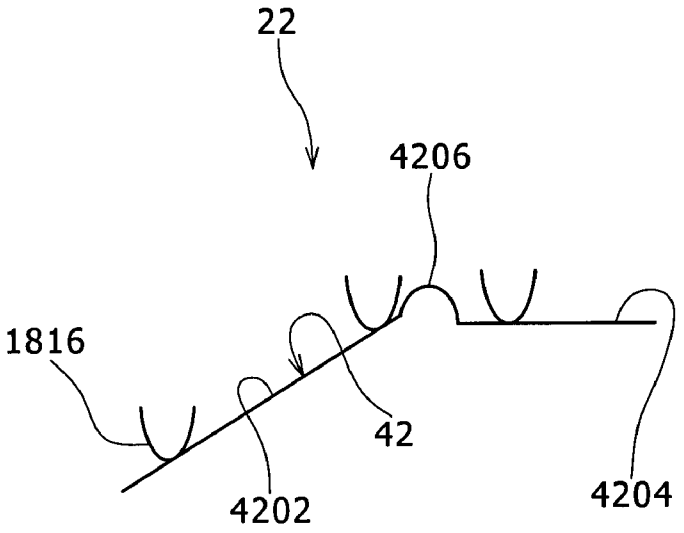


FIG. 11



SUCTION CUP DEVICE

CROSS REFERENCES TO RELATED APPLICATIONS

The present invention contains subject matter related to Japanese Patent Application JP 2006-305024, filed in the Japan Patent Office on Nov. 10, 2006, the entire contents of which being incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

This invention relates to a suction cup device.

2. Description of the Related Art

A suction cup device is used to attach various articles to an attaching object face.

Most of suction cup devices of the type described include a suction cup, a center shaft provided in a projecting manner from a rear face of the suction cup, a skirt member for covering the rear face of the suction cup, and an attaching base projecting from the center of the skirt member and having an outer tubular portion in which the center shaft is accommodated.

In order to stabilize an attached state of the suction cup device to the attaching object face, the suction face of the suction cup is opposed to the attaching object face to press the center shaft against the attaching object face such that the suction face is contacted over an overall face thereof with the attaching object face. Further, a rock lever is provided which displaces, in an initially attached state wherein an end of the skirt member contacts with an outer circumferential portion of the rear face of the suction cup, the center shaft in a direction away from the attaching object face to establish a finally attached state wherein the central portion of the suction face is spaced away from the attaching object face.

A suction cup device of the type described is disclosed, for example, in Japanese Utility Model Laid-Open No. Hei 4-133773.

SUMMARY OF THE INVENTION

However, in the existing suction cup device described above, since the rock lever rocks in a plane passing the center shaft, or in other words, since the rock lever rocks in a thicknesswise direction of the suction cup, it is necessary to assure a great space for the rocking motion of the rock lever. This is disadvantageous where it is tried to install a suction cup device at a narrow place.

Therefore, it is demanded to provide a suction cup device which eliminates the necessity to assure a large space for rocking motion of a rock lever and is advantageous in installation at a narrow plate.

According to an embodiment of the present invention, there is provided a suction cup device. The device includes a suction cup having a suction face and a rear face along a thicknesswise direction thereof, a center shaft provided at the center of the rear face of the suction cup and extending in a direction away from the rear face along the thicknesswise direction, and an attaching base having a skirt portion configured to cover the rear face of the suction cup and an outer tubular portion projecting from the center of the skirt portion and having the center shaft accommodated therein. The device further includes a rock lever coupled to an end portion of the center shaft against movement in an axial direction of the center shaft but for rotation in a circumferential direction of the center shaft between an initial position and a rocked position, and a coil spring disposed around the center shaft on the inner side of the outer tubular portion and anchored at one end thereof by the outer tubular portion and at the other end thereof by the rock lever in such a manner as to bias the rock lever toward the initial position and bias the outer tubular portion and the rock lever in a direction away from each other along the center shaft. The device still further includes a cam mechanism provided between the outer tubular portion and the rock lever and configured to move, in response to the rotation of the rock lever from the initial position to the rocked position, the suction cup from an initially attached state which is established by pressing, with the suction face opposed to an attaching object face, the center shaft against the attaching object face until the suction face is contacted over an overall area thereof with the attaching object face and an end portion of the skirt portion is contacted with an outer circumferential portion of the rear face or with a portion of the attaching object face on the outer side in a radial direction of the rear face to a finally attached state wherein the center shaft is displaced in a direction away from the attaching object face and then retain the rock lever at the rocked position against the biasing force of the coil spring.

According to another embodiment of the present invention, there is provided a suction cup device. The device includes a suction cup having a suction face and a rear face along a thicknesswise direction thereof, a center shaft provided at the center of the rear face of the suction cup and extending in a direction away from the rear face along the thicknesswise direction, and an attaching base having a skirt portion configured to cover the rear face of the suction cup and an outer tubular portion projecting from the center of the skirt portion and having the center shaft accommodated therein. The device further includes a rock lever coupled to an upper portion of the center shaft against movement in an axial direction of the center shaft but for rotation in a circumferential direction of the center shaft between an initial position and a rocked position, and a coil spring disposed around the center shaft on the inner side of the outer tubular portion and anchored at one end thereof by the outer tubular portion and at the other end thereof by the rock lever in such a manner as to bias the rock lever toward the initial position and bias the outer tubular portion and the rock lever in a direction away from each other along the center shaft. The suction cup has an intermediately attached state in which the center shaft is displaced in a direction away from an attaching object face from a position thereof in an initially attached state of the suction cup device in which the suction face contacts over an overall area thereof with the attaching object face with the suction face opposed to the attaching object face and with the center shaft pressed against the attaching object face and a lower end of the skirt portion is contacted with an outer circumferential portion of the rear face or with a portion of the attaching object face on the outer side of the rear face in a radial direction being formed by the coil spring. The suction cup device has a cam mechanism configured to move, in response to rotation of the rock lever from the initial position to the rocked position, the suction cup from the position in the intermediately attached state to another position in a finally attached state in which the center shaft is displaced further in the direction away from the attaching object face and then retain the rock lever at the rocked position against the biasing force of the coil spring.

With the suction cup devices, the finally attached state of the suction cup device can be established by rotating the rock lever around the center shaft. Consequently, the suction cup devices can be attached to a small space advantageously.

The features and advantages of the present invention will become apparent from the following description and the appended claims, taken in conjunction with the accompanying drawings in which like parts or elements denoted by like reference symbols.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a sectional view showing a suction cup device to which the present invention is applied and which is in a non-attached state;

FIG. **2** is a sectional view of the suction cup device in an initially attached state;

FIG. **3** is a sectional view of the suction cup device in an intermediately attached state;

FIG. **4** is a sectional view of the suction cup device in a finally attached state;

FIGS. **5**A and **5**B are perspective views showing the suction cup device in the intermediately attached state and the finally attached state, respectively;

FIG. **6** is a perspective view of a suction cup of the suction cup device;

FIG. **7** is a cross sectional view of the suction cup;

FIG. **8** is a perspective view of an attaching base of the suction cup device;

FIGS. **9**A and **9**B are perspective views of a second member of a rock lever of the suction cup device and FIGS. **9**C and **9**D are perspective views of a first member of the rock lever;

FIG. **10** is a perspective view of a coil spring of the suction cup device; and

FIG. **11** is a schematic view illustrating operation of a cam mechanism of the suction cup device.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

Referring first to FIG. **1**, there is shown a suction cup device **10** to which the present invention is applied. The suction cup device **10** is used to support a display panel of a car navigation system or a television apparatus on the dashboard in the cabin of an automobile or support a controller for remotely controlling various vehicle-carried apparatus or a like apparatus.

As shown in FIG. **1**, the suction cup device **10** includes a suction cup **12**, a center shaft **14**, an attaching base **16**, a rock lever **18**, a coil spring **20**, a cam mechanism **22** and so forth.

Referring to FIGS. **1** and **6**, the suction cup **12** is generally in the form of a disk having two faces in the thicknesswise direction one of which is formed as a concave suction face **24** and the other of which is formed as a convex rear face **26**.

The suction cup **12** can be made of various hitherto known resilient materials hitherto used for suction cups such as, for example, urethane type materials, styrene type materials and silicon resin materials.

It is to be noted that a piece portion **12**A for removing operation is provided on an outer edge of the suction cup **12** such that it extends in a diametrically outward direction.

Further, the suction cup **12** can be formed in various hitherto known configurations.

Referring now to FIGS. **1** and **7**, for example, the suction cup **12** may include a suction cup body **28** in the form of a disk which forms the rear face **26** and a gel layer **30** which forms the suction face **24**.

The suction cup body **28** can be made of synthetic resin materials having resiliency, for example, various hitherto known resilient materials hitherto used for suction cups such as urethane type materials, styrene type materials and silicon resin materials similarly to the suction cup **12**.

The gel layer **30** is made of gel and bonded to a concave attaching face **2802**, which is formed on a face of the suction cup body **28** positioned remotely from the rear face **26**, in such a manner as to cover the attaching face **2802**. The suction face **24** is formed from the surface of the gel layer **30**.

For such gel, gel of a synthetic resin type such as a polyethylene type, a styrene type and the silicon resin type can be used. Further, as the gel of the polyethylene type, for example, "COSMO SUPER GEL" by Cosmo Instruments Co., Ltd. which is an article on the market can be used. As the gel of the styrene type, for example, "NAGFLEX" by Inoac Corporation which is an article on the market can be used. As the gel of the silicon type, for example, "αGEL" (alpha gel) by GELTEC Co., Ltd. which is an article on the market can be used.

Further, for the bonding of the gel layer **30** to the attaching face **2802**, such a method of bonding by two-color molding or by means of a bonding agent can be adopted.

Where the suction face **24** is formed from the gel layer **30**, even if the attaching object face **2** is formed as a leather embossed face which has fine convexes and concaves or provides a coarse feel, the suction face **24** is deformed following up the convex and concave geometry. Consequently, the suction face **24** can contact closely with the convex and concave face or coarse face without allowing a gap formed therebetween, thereby attaching the suction cup **12** steadily on those kinds of faces. This is advantageous in that various articles are attached with certainty to an attaching object face **2** such as a leather embossed face of the dashboard panel or the like.

Referring to FIG. **1**, the center shaft **14** extends in a direction away from the rear face **26** along the thicknesswise direction of the suction cup **12**, that is, along the center shaft of the suction face **24**, from a central portion of the rear face **26** of the suction cup **12**.

The center shaft **14** is formed from a material having rigidity such as, for example, a metal or a hard synthetic resin and is embedded at a lower end thereof in a swollen portion at the center of the suction cup **12**.

The attaching base **16** supports such a display panel, a controller or the like as described above on the suction cup **12**.

The attaching base **16** is formed from a material having rigidity such as a hard synthetic resin material.

The attaching base **16** has a skirt portion **32**, an outer tubular portion **34**, an attaching portion **36** and so forth as seen in FIG. **8**.

The skirt portion **32** has a spherical surface shape and covers the rear face **26** of the suction cup **12**.

Referring to FIG. **1**, a resilient member **38** is provided along an overall circumference of an end of the skirt portion **32**.

The resilient member **38** is formed from a resilient material having a hardness lower than that of the resilient material of the suction cup **12** such that it can be deformed readily in accordance with the shape of the attaching object face **2**. In particular, even if the attaching object face **2** has a non-planar shape such as a spherical shape or a cylindrical shape, the resilient member **38** is deformed following up the shape of the attaching object face **2**. Consequently, the suction cup **12** is advantageously allowed to be attached with certainty to the attaching object face **2** of such non-planar shape.

More particularly, the suction cup **12** is formed from a resilient material having a JIS-A hardness of 70 or more while the resilient member **38** is formed from a resilient material of another JIS-A hardness of 40 or more but less than 70.

As such resilient materials, typically a polyurethane foam material can be used, and for example, "PORON" by Inoac Corporation which is an article on the market can be used.

The resilient member **38** has a thickness in a direction parallel to the thicknesswise direction of the suction cup **12** and has an annular belt-like plate shape centered at the center of the suction cup **12**, that is, centered at the center of the skirt portion **32**.

The resilient member **38** is a portion of the skirt portion **32** which is contacted by the rear face **26** of the attaching object face **2** in an initially attached state, an intermediately attached state and a finally attached state hereinafter described. It is to be noted that the resilient member **38** may be omitted such that the end of the skirt portion **32** is contacted directly with the outer circumference of the rear face **26** of the suction cup **12** or the end of the skirt portion **32** is contacted with a portion of the attaching object face **2** which is positioned on the outer side of the suction cup **12** as indicated by reference character **2**A in FIG. **2**.

The outer tubular portion **34** is erected uprightly from the center of the skirt portion **32** and has the center shaft **14** accommodated on the inner side thereof.

Referring to FIGS. **1** and **8**, an intermediate wall **40** is provided on the inner circumference of an intermediate portion in an axial direction of the outer tubular portion **34** such that it projects in a radially inward direction.

Meanwhile, two cams **42** are provided at portions of the inner circumference of the outer tubular portion **34** positioned on the intermediate wall **40** and spaced by 180 degrees from each other such that they extend along a circumferential direction of the inner circumference of the outer tubular portion **34**.

Referring to FIGS. **8** and **11**, each of the cams **42** includes an inclined cam **4202**, an upper cam **4204** and a locking projection **4206**. The inclined cam **4202** extends along a circumferential direction on the inner circumference of the outer tubular portion **34** while being displaced in a direction toward an end of the center shaft **14**. The upper cam **4204** connects to an upper end of the inclined cam **4202** displaced most in the direction toward the end of the center shaft **14** and extends along the circumferential direction on the inner circumference of the outer tubular portion **34** at a height substantially equal to the upper end of the inclined cam **4202**. The locking projection **4206** is provided at a portion of the upper cam **4204** rather near to the inclined cam **4202**.

Referring to FIG. **8**, a cutaway portion **4210** is formed at an end portion in an axial direction of the outer tubular portion **34** remote from the suction cup **12** such that it extends along the circumferential direction of the outer tubular portion **34**. The cutaway portion **4210** is provided to allow the rock lever **18** to rotate between an initial position and a rocked position. More particularly, a lever portion **46** hereinafter described is arrested by one end of the cutaway portion **4210** in the extension direction to define the initial position, and is arrested by the other end of the cutaway portion **4210** in the extension direction to define the rocked position.

The attaching portion **36** is provided at an end portion in the axial direction of the outer tubular portion **34** remote from the suction cup **12** as seen in FIG. **1**.

The attaching portion **36** is a portion to which an article to be supported on the attaching object face **2** is attached by the suction cup device **10**. In the present embodiment, the article is, for example, a controller for remotely controlling a display panel, various vehicle-carried apparatus or a like apparatus.

The rock lever **18** is coupled to an end portion of the center shaft **14** against movement in an axial direction of the center shaft **14** but for rotation between such an initial position as seen in FIG. **5**A and such a rocked position as seen in FIG. **5**B in a circumferential direction of the center shaft **14**.

The rock lever **18** is formed from a material having rigidity such as a hard synthetic resin material.

The rock lever **18** has an inner tubular portion **44** inserted between the outer circumference of the center shaft **14** and the inner circumference of the outer tubular portion **34** and coupled to the center shaft **14** against movement in the axial direction of the center shaft **14** but for rotation around the center shaft **14**. The rock lever **18** further has a lever portion **46** for rotating operation extending outwardly in a radial direction of the inner tubular portion **44** from the inner tubular portion **44**.

In the present embodiment, the rock lever **18** is formed from two members including a first member **18**A and a second member **18***b* as seen in FIGS. **9**A to **9**D.

The first member **18**A has a tubular portion **1802** and a first end wall **1804** which closes up one end in an axial direction of the tubular portion **1802**. The lever portion **46** is provided in a projecting manner on the tubular portion **1802**.

The second member **18**B has a tubular portion **1812**, and a second end wall **1814** which closes up one end in an axial direction of the tubular portion **1812**. A pair of projections **1816** are provided at different portions, spaced by a distance of 180 degrees from each other, at the other end portion in the axial direction of the tubular portion **1812** positioned rather near to the suction cup **12**. A hole **1818** is formed at the center of the second end wall **1814**.

In the present embodiment, the cam mechanism **22** is formed from the cams **42** of the outer tubular portion **34** and the projections **1816** of the second member **18**B as seen in FIG. **11**.

Assembly of the first member **18**A and the second member **18***b* is performed in the following manner.

First, the tubular portion **1812** of the second member **18***b* is inserted between the outer circumference of the center shaft **14** and the inner circumference of the outer tubular portion **34** as seen in FIGS. **1** and **9**A to **9**D. Then, an end portion of the center shaft **14** is inserted into the hole **1818**, and a stop ring **1820** is attached to a portion of the center shaft **14** which projects from the hole **1818** thereby to attach the second member **18***b* to the end portion of the center shaft **14** against movement in the axial direction but for rotation in a circumferential direction of the center shaft **14**.

Then, the first member **18**A is fitted to the outer side with and fixed to the second member **18**B.

This fitting fixation is performed such that projections **1822** (refer to FIG. **9**B) of the tubular portion **1812** of the second member **18**B are fitted into cutaway portion **1824** (refer to FIG. **9**D) of the tubular portion **1802** of the first member **18**A and a projection **1826** (refer to FIG. **9**D) on the inner circumference of the tubular portion **1802** of the first member **18**A is fitted into a recess **1828** (refer to FIG. **9**A) on the outer circumference of the tubular portion **1812** of the second member **18**B.

Accordingly, in the present embodiment, the inner tubular portion **44** (refer to FIG. **1**) is formed from the tubular portion **1802** of the first member **18**A and the tubular portion **1812** of the second member **18**B.

Then, the first and second members **18**A and **18**B are attached to the center shaft **14** such that the lever portion **46** is positioned above the cutaway portion **4210** (refer to FIG. **8**). More particularly, the first and second members **18**A and **18**B are attached to the center shaft **14** such that the projections **1816** are positioned between the two cams **42** while the lever

portion **46** is positioned at the initial position at one end portion of the cutaway portion **4210** in the extension direction.

The first end wall **1804** and the second end wall **1814** are positioned closely to each other in a state wherein the first and second members **18**A and **18**B are fitted on the center shaft **14**, and the first end wall **1804** is provided at an end portion in the axial direction of the inner tubular portion **44** positioned rather near to the end portion of the center shaft **14** such that it closes up the end portion of the inner tubular portion **44**. Further, the second end wall **1814** is provided at a portion in the inside of the inner tubular portion **44** rather near to the first end wall **1804** such that it projects inwardly in a radial direction of the inner tubular portion **44**.

Referring to FIG. **10**, the coil spring **20** has a coiled portion **20**A, and a pair of end portions **20**B and **20**C bent at the opposite ends of the coiled portion **20**A.

Referring to FIG. **1**, the coiled portion **20**A of the coil spring **20** is disposed around the center shaft **14** on the inner side of the outer tubular portion **34**, and the end portion **20**B is anchored by the outer tubular portion **34** while the end portion **20**C is anchored by the rock lever **18** to bias the rock lever **18** toward the initial position and bias the outer tubular portion **34** and the rock lever **18** to move away from each other along the center shaft **14**.

More particularly, the coil spring **20** is disposed between the second end wall **1814** and the intermediate wall **40** around the center shaft **14** as seen in FIG. **2**. The coil spring **20** is anchored at the end portion **20**B thereof by the intermediate wall **40** and at the end portion **20**C thereof by the second end wall **1814** and biases the rock lever **18** toward the initial position while it biases the outer tubular portion **34** and the center shaft **14** to move away from each other.

In the present embodiment, the coil spring **20** is provided to exert resilient force for obtaining an intermediately attached position hereinafter described.

Now, operation of the suction cup device **10** is described.

If the suction face **24** of the suction cup **12** is opposed to the attaching object face **2** and the center shaft **14** is pressed against the attaching object face **2** until the suction face **24** is contacted over an overall area thereof with the attaching object face **2** and the end of the skirt portion **32** is contacted with an outer circumferential portion of the rear face **26** or with a portion (refer to **2**A in FIG. **2**) of the attaching object face **2** on the outer side in a radial direction of the rear face **26**. The initially attached state illustrated in FIGS. **2** and **5**A is established thereby. It is to be noted that the operation of pressing the center shaft **14** against the attaching object face **2** is performed by a user pressing the first end wall **1804** of the rock lever **18** with its finger.

If the rock lever **18** is rotated from the initial position to the rocked position while the suction cup device **10** is in the initially attached state described above, then the center shaft **14** is displaced compulsorily in a direction away from the attaching object face **2** by the cam mechanism **22** thereby to establish the finally attached state illustrated in FIGS. **4** and **5**B.

More specifically, if the rock lever **18** is rotated from the initial position to the rocked position, then the projections **1816** slidably move on the inclined cams **4202** as seen in FIG. **11**. Consequently, the center shaft **14** is displaced compulsorily in a direction away from the attaching object face **2**. The projections **1816** soon ride over the locking projections **4206** and come to the upper cams **4204**. While the rock lever **18** tends to return to the initial position by the biasing force of the coil spring **20**, since the projections **1816** engage with the locking projections **4206**, the rock lever **18** remains at the rocked position.

In the present embodiment, if the first end wall **1804** of the rock lever **18** is released from the finger while the suction cup device **10** is in the initially attached state, then the rock lever **18** is moved to the intermediately attached position illustrated in FIG. **3** by the coil spring **20**.

The intermediately attached position is established by the center shaft **14** being displaced in a direction away from the attaching object face **2** by the resilient force of the coil spring **20** in the initially attached state.

Accordingly, in the present embodiment, the finally attached state is established when, while the suction cup device **10** is in the intermediately attached state, the rock lever **18** is rotated from the initial position to the rocked position to further displace the center shaft **14** in the direction away from the attaching object face **2**.

The suction cup device **10** is attached firmly to the attaching object face **2** in such a finally attached state as just described, and such an article as a controller for remotely controlling a display panel or various vehicle-carried articles is supported on the dashboard in the cabin of an automobile.

On the other hand, removal of the suction cup device **10** from the attaching object face **2** is performed by exfoliating the piece portion **12**A from the attaching object face **2** to exfoliate the suction cup **12**.

If the suction cup **12** is removed from the attaching object face **2**, then the outer tubular portion **34** and the rock lever **18** are spaced away from each other along the center shaft **14** by the coil spring **20**, and consequently, the projections **1816** are displaced from the cams **42**. Then, the rock lever **18** is rotated from the rocked position toward the initial position by the biasing force of the coil spring **20** until it is returned to the initial position.

Accordingly, with the suction cup device **10** of the present embodiment, the finally attached state can be established by an operation of the rock lever **18**, which is advantageous when it is tried to firmly attach the suction cup device **10** to the attaching object face **2**.

Further, since the operation of the rock lever **18** is performed by rotating the rock lever **18** around the center shaft **14**, the rock lever **18** can be operated to rotate without assuring a great space. Accordingly, the suction cup device **10** can be attached advantageously.

Further, if the suction cup device **10** is removed from the attaching object face **2**, then the rock lever **18** automatically returns to the initial position. Consequently, the operability of the suction cup device **10** can be enhanced advantageously.

Further, although the intermediately attached state may be omitted, where the intermediately attached state is implemented using the coil spring **20** as in the embodiment described above, then the suction cup device **10** can be attached firmly to the attaching object face **2** advantageously by the finally attached state.

Further, with the suction cup device **10** of the present embodiment, the action of biasing the rock lever **18** from the rocked position to the initial position and the action of biasing the outer tubular portion **34** and the rock lever **18** in a direction away from each other along the center shaft **14** are implemented by the single coil spring **20**. Therefore, reduction of the number of parts and simplification of the structure can be anticipated advantageously.

It is to be noted that, while, in the embodiment described above, the suction cup device **10** is used to attach a controller for a display panel or various vehicle-carried apparatus, the article to be supported by the suction cup device **10** is not limited to such vehicle-carried articles, but the present invention can be applied to various other apparatus.

While a preferred embodiment of the present invention has been described using specific terms, such description is for illustrative purpose only, and it is to be understood that changes and variations may be made without departing from the spirit or scope of the following claims.

What is claimed is:

1. A suction cup device, comprising:

a suction cup having a suction face and a rear face along a thicknesswise direction thereof;

a center shaft provided at the center of said rear face of said suction cup and extending in a direction away from said rear face along the thicknesswise direction;

an attaching base having a skirt portion configured to cover said rear face of said suction cup and an outer tubular portion projecting from the center of said skirt portion and having said center shaft accommodated therein;

a rock lever coupled to an end portion of said center shaft against movement in an axial direction of said center shaft but for rotation in a circumferential direction of said center shaft between an initial position and a rocked position;

a coil spring disposed around said center shaft on the inner side of said outer tubular portion and anchored at one end thereof by said outer tubular portion and at the other end thereof by said rock lever in such a manner as to bias said rock lever toward the initial position and bias said outer tubular portion and said rock lever in a direction away from each other along said center shaft; and

a cam mechanism provided between said outer tubular portion and said rock lever and configured to move, in response to the rotation of said rock lever from the initial position to the rocked position, said suction cup from an initially attached state which is established by pressing, with said suction face opposed to an attaching object face, said center shaft against the attaching object face until said suction face is contacted over an overall area thereof with the attaching object face and an end portion of said skirt portion is contacted with an outer circumferential portion of said rear face or with a portion of the attaching object face on the outer side in a radial direction of said rear face to a finally attached state wherein said center shaft is displaced in a direction away from the attaching object face and then retain said rock lever at the rocked position against the biasing force of said coil spring.

2. The suction cup device according to claim 1, wherein:

said suction cup further has an intermediately attached state wherein said center shaft is displaced in the direction away from the attaching object face by said coil spring from the position thereof when said suction cup is in the initially attached state; and

the finally attached state is established by rotating, while said suction cup is in the intermediately attached state, said rock lever from the initial position to the rocked position to further displace said center shaft in the direction away from the attaching object face.

3. The suction cup device according to claim 1, wherein:

said rock lever has an inner tubular portion inserted between the outer circumference of said center shaft and the inner circumference of said outer tubular portion and coupled to said center shaft against movement in the axial direction of said center shaft but for rotation around said center shaft, and a lever portion for rotating operation projecting outwardly in a radial direction of said inner tubular portion from said inner tubular portion; and

said cam mechanism is provided between the inner circumference of said outer tubular portion and said inner tubular portion.

4. The suction cup device according to claim 1, wherein:

said rock lever has an inner tubular portion inserted between the outer circumference of said center shaft and the inner circumference of said outer tubular portion and coupled to said center shaft against movement in the axial direction of said center shaft but for rotation around said center shaft, and a lever portion projecting outwardly in a radial direction of said inner tubular portion from said inner tubular portion; and

said cam mechanism includes a cam provided on the inner circumference of an intermediate portion in the axis direction of said outer tubular portion and extending in a circumferential direction of the inner circumference, and a projection provided at an end portion in the axial direction of said inner tubular portion positioned rather near to said suction cup and configured to engage with said cam.

5. The suction cup device according to claim 1, wherein:

said rock lever has an inner tubular portion inserted between the outer circumference of said center shaft and the inner circumference of said outer tubular portion and coupled to said center shaft against movement in the axial direction of said center shaft but for rotation around said center shaft, and a lever portion projecting outwardly in a radial direction of said inner tubular portion from said inner tubular portion;

an intermediate wall is provided on the inner circumference of an intermediate portion in the axial direction of said outer tubular portion in such a manner as to be positioned in the proximity of or contacted with an end portion in the axial direction of said inner tubular portion positioned rather near to said suction cup; and

said cam mechanism includes a cam provided on said intermediate wall and extending in a circumferential direction of said intermediate wall, and a projection provided at an end portion in the axial direction of said inner tubular portion positioned rather near to said suction cup and configured to engage with said cam.

6. The suction cup device according to claim 1, wherein:

said rock lever has an inner tubular portion inserted between the outer circumference of said center shaft and the inner circumference of said outer tubular portion and coupled to said center shaft against movement in the axial direction of said center shaft but for rotation around said center shaft, and a lever portion projecting outwardly in a radial direction of said inner tubular portion from said inner tubular portion;

a first end wall being provided at an end portion in the axial direction of said inner tubular position positioned rather near to the end portion of said center shaft in such a manner as to close up the end portion of said inner tubular portion,

a second end wall being provided at a portion in the inside of said inner tubular portion rather near to said first end face in such a manner as to project inwardly in a radial direction of said inner tubular portion,

said first end wall serving as a pressing portion for pressing said center shaft against the attaching object face when the initially attached state is to be established,

said second end wall being coupled to an end portion of said center shaft against movement in the axial direction of said center shaft but for rotation around said center shaft,

an intermediate wall being provided on the inner circumference of an intermediate portion in the axial direction of said outer tubular portion in such a manner as to project inwardly in a radial direction, and

said coil spring being disposed between said second end wall and said intermediate wall around said center shaft.

7. The suction cup device according to claim 1, wherein:
said rock lever has an inner tubular portion inserted between the outer circumference of said center shaft and the inner circumference of said outer tubular portion and coupled to said center shaft against movement in the axial direction of said center shaft but for rotation around said center shaft, and a lever portion projecting outwardly in a radial direction of said inner tubular portion from said inner tubular portion;
a first end wall being provided at an end portion in the axial direction of said inner tubular position positioned rather near to the end portion of said center shaft in such a manner as to close up the end portion of said inner tubular portion,
a second end wall being provided at a portion in the inside of said inner tubular portion rather near to said first end face in such a manner as to project inwardly in a radial direction of said inner tubular portion,
said first end wall serving as a pressing portion for pressing said center shaft against the attaching object face when the initially attached state is to be established,
said second end wall being coupled to an end portion of said center shaft against movement in the axial direction of said center shaft but for rotation around said center shaft,
an intermediate wall being provided on the inner circumference of an intermediate portion in the axial direction of said outer tubular portion in such a manner as to project inwardly in a radial direction,
said coil spring being disposed between said second end wall and said intermediate wall around said center shaft,
said coil spring being anchored at one end thereof by said intermediate wall and at the other end thereof by said end wall.

8. The suction cup device according to claim 1, wherein:
said rock lever has an inner tubular portion inserted between the outer circumference of said center shaft and the inner circumference of said outer tubular portion and coupled to said center shaft against movement in the axial direction of said center shaft but for rotation around said center shaft, and a lever portion projecting outwardly in a radial direction of said inner tubular portion from said inner tubular portion;
said cam mechanism includes a cam provided on the inner circumference of an intermediate portion in the axis direction of said outer tubular portion and extending in a circumferential direction of the inner circumference, and a projection provided at an end portion in the axial direction of said inner tubular portion positioned rather near to said suction cup and configured to engage with said cam; and
said cam includes an inclined cam extending along a circumferential direction of the inner circumference of said outer tubular portion while being displaced in a direction toward an end portion of said center shaft, an upper cam connecting to an end portion of said inclined cam displaced most in the direction toward the end portion of said center shaft and extending along a circumferential direction of the circumference of said outer tubular portion, and an anchoring projection provided at a portion of said upper cam rather near to said inclined cam.

9. The suction cup device according to claim 1, wherein:
said rock lever has an inner tubular portion inserted between the outer circumference of said center shaft and the inner circumference of said outer tubular portion and coupled to said center shaft against movement in the axial direction of said center shaft but for rotation around said center shaft, and a lever portion for rotating operation projecting outwardly in a radial direction of said inner tubular portion from said inner tubular portion;
a cutaway portion being formed on said outer tubular portion so as to extend in the circumferential direction such that said lever is inserted into said cutaway portion,
said lever portion being locked by one end of said cutaway portion in the extension direction to define the initial position whereas said lever portion is locked by the other end of said cutaway portion in the extension direction to define the rocked position.

10. The suction cup device according to claim 1, wherein:
said suction cup includes a suction cup body and a gel layer;
said suction cup body being formed from a resilient material while forming said rear face,
said gel layer being formed from gel and attached to said suction cup body to form said suction face.

11. A suction cup device, comprising:
a suction cup having a suction face and a rear face along a thicknesswise direction thereof;
a center shaft provided at the center of said rear face of said suction cup and extending in a direction away from said rear face along the thicknesswise direction;
an attaching base having a skirt portion configured to cover said rear face of said suction cup and an outer tubular portion projecting from the center of said skirt portion and having said center shaft accommodated therein;
a rock lever coupled to an upper portion of said center shaft against movement in an axial direction of said center shaft but for rotation in a circumferential direction of said center shaft between an initial position and a rocked position; and
a coil spring disposed around said center shaft on the inner side of said outer tubular portion and anchored at one end thereof by said outer tubular portion and at the other end thereof by said rock lever in such a manner as to bias said rock lever toward the initial position and bias said outer tubular portion and said rock lever in a direction away from each other along said center shaft; wherein
said suction cup has an intermediately attached state in which said center shaft is displaced in a direction away from an attaching object face from a position thereof in an initially attached state of said suction cup device in which said suction face contacts over an overall area thereof with the attaching object face with said suction face opposed to the attaching object face and with said center shaft pressed against the attaching object face and a lower end of said skirt portion is contacted with an outer circumferential portion of said rear face or with a portion of the attaching object face on the outer side of said rear face in a radial direction being formed by said coil spring,
said suction cup device has a cam mechanism configured to move, in response to rotation of said rock lever from the initial position to the rocked position, said suction cup from the position in the intermediately attached state to another position in a finally attached state in which said center shaft is displaced further in the direction away from the attaching object face and then retain said rock lever at the rocked position against the biasing force of said coil spring.

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